

Probability and Statistics – for Computer Science

Based on Forsyth, *Probability and Statistics for Computer Science* (2018)

Applied Statistics and Probability for Computer Scientists

First Tools for Looking at Data

Chapter 1 — Datasets, Visualization, and Summary Statistics

Dr. Khalil Chebil

Probability & Statistics for Computer Science

Chapter Outline | What we will cover

What Is a Dataset?
Visualizing Data
Measures of Central Tendency
Measures of Spread

Standard Coordinates & Box Plots
Case Study: Australian Pizzas
Summary

Learning Objectives

- ▶ Classify data as **categorical**, **ordinal**, **discrete**, or **continuous**
- ▶ Build and read **bar charts** and **histograms**
- ▶ Compute **mean**, **median**, **variance**, and **IQR**
- ▶ Use **standard coordinates** and **box plots** to compare datasets

1

Section 1

What Is a Dataset?

What Is a Dataset? | A collection of measurements, described as d -tuples

Definition

A **dataset** $\{x\}$ is a collection of N items; each item is a **d -tuple** of measurements, $x_i = (x_i^1, x_i^2, \dots, x_i^d)$.

- ▶ N — number of items (rows)
- ▶ d — number of attributes (columns)
- ▶ Each attribute is a **dimension** of the tuple

Students dataset ($d = 3$)

Name	Age	GPA
Alice	21	3.7
Bob	23	3.2
Cara	22	3.9

$N = 3$ items, $d = 3$ attributes each

Types of Data | Choosing the right tool starts with the right type

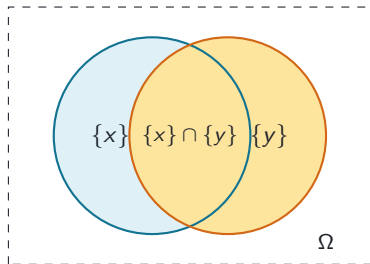
Type	Meaning	Examples
Categorical	Labels, no order	Country, blood type, gender
Ordinal	Labels, natural order	Letter grade, satisfaction (low/med/high)
Discrete	Countable numeric values	# of siblings, # of goals scored
Continuous	Any value in a range	Height, temperature, time

Why it matters

The type of data determines which summaries and plots make sense — a mean of postal codes is meaningless, but a mean of temperatures is not.

Set Operations on Datasets | Datasets behave like sets of items

- ▶ **Union** $\{x\} \cup \{y\}$ — items in either
- ▶ **Intersection** $\{x\} \cap \{y\}$ — items in both
- ▶ **Difference** $\{x\} \setminus \{y\}$ — in $\{x\}$ only
- ▶ **Complement** $\{x\}^c$ — relative to universe Ω
- ▶ **Disjoint** — $\{x\} \cap \{y\} = \emptyset$





Section 2

Visualizing Data

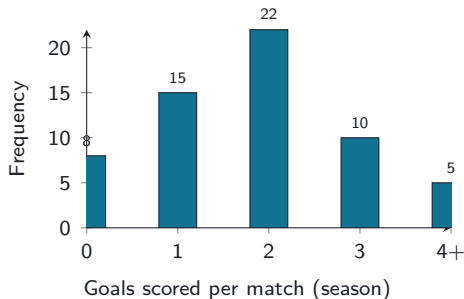
Bar Charts | Comparing counts across categories

Definition

A bar chart shows the **frequency** of each category as a bar; bar *height* = count (or proportion).

Use when

Data is **categorical** or **ordinal** — order and gaps between bars carry no numeric meaning.



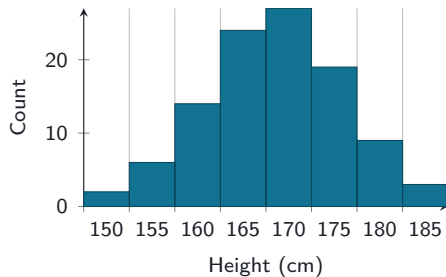
Histograms | The workhorse plot for numeric data

Definition

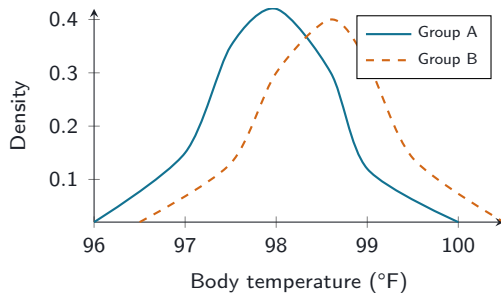
Split the range into **bins**; bar height = count of values falling in that bin.

Choosing bin count

- ▶ Too few bins $\Rightarrow$ hides structure
- ▶ Too many bins $\Rightarrow$ noisy, sparse
- ▶ Rule of thumb: $\sqrt{N}$ or Sturges' rule



Conditional Histograms | Splitting by a categorical variable reveals structure



Why compare conditionally?

- ▶ Reveals **subgroup** differences hidden in the pooled data
- ▶ Same range, different **shape** or **center**
- ▶ Key insight: *overlap* vs. *separation*

Section 3

Measures of Central Tendency

Definition

$$\text{mean}(\{x\}) = \bar{x} = \frac{1}{N} \sum_{i=1}^N x_i$$

Properties

- ▶ Translation: $\text{mean}(\{x + c\}) = \bar{x} + c$
- ▶ Scaling: $\text{mean}(\{kx\}) = k\bar{x}$
- ▶ Minimizes $\sum_i (x_i - v)^2$ over v
- ▶ **Sensitive** to outliers

After seeing k items, let $\hat{\mu}_k, \hat{\sigma}_k$ be the running estimates. On seeing x_{k+1} :

Mean update

$$\hat{\mu}_{k+1} = \frac{k \hat{\mu}_k + x_{k+1}}{k + 1}$$

Std. dev. update

$$\hat{\sigma}_{k+1} = \sqrt{\frac{k \hat{\sigma}_k^2 + (x_{k+1} - \hat{\mu}_{k+1})^2}{k + 1}}$$

Why it matters

- ▶ No need to store all N values
- ▶ Constant memory, constant work per item
- ▶ Ideal for **streaming** sensor / log data

$\mathcal{O}(1)$ time and space per update

The Median | The middle value, robust to outliers

Definition

Sort $\{x\}$; the median is the middle value (N odd) or the average of the two middle values (N even).

- ▶ **Robust**: unaffected by extreme outliers
- ▶ Splits data into two equal halves

Median vs. mean with outliers

Salaries: $\{30, 32, 35, 33, 500\}$ (in \$k)

$$\text{mean} = 126$$

$$\text{median} = 33$$

The median better reflects a **typical** salary.

4

Section 4

Measures of Spread

Variance & Standard Deviation | How far, on average, from the mean?

Definition

$$\text{var}(\{x\}) = \sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2$$
$$\text{std}(\{x\}) = \sigma = \sqrt{\text{var}(\{x\})}$$

Properties

- ▶ Translation invariant: $\text{std}(\{x + c\}) = \sigma$
- ▶ Scaling: $\text{std}(\{kx\}) = |k| \sigma$
- ▶ Same **units** as the data (unlike variance)

Percentiles & quartiles

- ▶ $\text{percentile}(\{x\}, k)$: value below which $k\%$ of data lie
- ▶ $Q1 = \text{percentile}(\{x\}, 25)$
- ▶ $Q2 = \text{percentile}(\{x\}, 50) = \text{median}$
- ▶ $Q3 = \text{percentile}(\{x\}, 75)$

Interquartile range

$$\text{IQR}(\{x\}) = Q3 - Q1$$

Robust measure of the spread of the **middle 50%**.

- ▶ Translation invariant:
 $\text{IQR}(\{x + c\}) = \text{IQR}(\{x\})$
- ▶ Scaling: $\text{IQR}(\{kx\}) = |k| \text{IQR}(\{x\})$

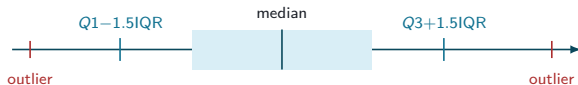
Detecting Outliers with the IQR | The rule behind every box-plot whisker

The $1.5 \times \text{IQR}$ rule

An item x_i is an outlier if

$$x_i < Q1 - 1.5 \text{IQR}$$

$$\text{or } x_i > Q3 + 1.5 \text{IQR}$$



Section 5

Standard Coordinates & Box Plots

Standard Coordinates (Z-Scores) | Comparing values across different scales

Definition

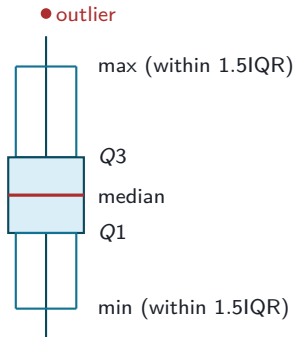
$$\hat{x}_i = \frac{x_i - \bar{x}}{\sigma}$$

- ▶ $\text{mean}(\{\hat{x}\}) = 0$ *always*
- ▶ $\text{std}(\{\hat{x}\}) = 1$ *always*
- ▶ **Unitless** — comparable across measurements

Reading a z-score

$\hat{x}$	Interpretation
0	exactly at the mean
+1	1 s.d. above mean
-2	2 s.d. below mean

Anatomy of a Box Plot | Five numbers, one picture



Components

- ▶ Box: $Q1$ to $Q3$ (the IQR)
- ▶ Line inside box: median
- ▶ Whiskers: extend to most extreme point within 1.5 IQR
- ▶ Dots: outliers beyond the whiskers

Using Summaries Sensibly | Which number to report, and how

Mean vs. median

- ▶ Symmetric, no outliers $\Rightarrow$ mean is fine
- ▶ Skewed or outlier-prone $\Rightarrow$ prefer median

Reporting precision

- ▶ Don't report more digits than the data supports
- ▶ Categorical data: report **mode**, not mean
- ▶ Always pair a **location** with a **spread**

Section 6

Case Study: Australian Pizzas

Case Study — Whose Pizza Is Bigger? | EagleBoys claims a bigger pizza than Dominos

The question

An advertising claim: EagleBoys' pizzas are bigger than Dominos'. Is it true, and by how much?

The dataset

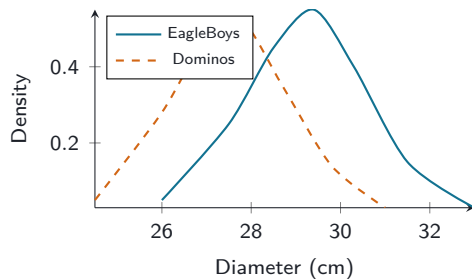
Diameters (cm) measured from a sample of pizzas ordered from each chain — a numeric, continuous, two-**group** comparison.

Method

Step 1: compute summary statistics for each brand. Step 2: visualize with histograms / box plots. Step 3: interpret *with caveats*.

Comparing the Two Brands | Summary statistics side by side

Statistic	EagleBoys	Dominos
Mean (cm)	29.4	27.8
Median (cm)	29.5	27.9
Std. dev.	1.1	1.3



Evidence for EagleBoys' claim

- ▶ Higher mean *and* median diameter
- ▶ Distributions barely overlap

Important caveats

- ▶ Small sample — is the difference **significant**?
- ▶ Sampling bias: which stores, which day?
- ▶ Bigger diameter $\neq$ more *value* (price, toppings)

The answer

On this sample, EagleBoys' pizzas *are* measurably larger — but a rigorous claim needs the hypothesis-testing tools of Part III.



Section 7

Summary

Location vs. Scale Parameters | The two questions every summary answers

Question	Statistic	Robust to outliers?
Where is the data? (location)	Mean	No
	Median	Yes
How spread out? (scale)	Std. dev. σ	No
	IQR	Yes

Describe

- ▶ Data types drive the choice of summary and plot
- ▶ Bar charts for categories, histograms for numbers

Summarize

Mean/median for location, std. dev./IQR for spread, z-scores to compare across scales.

Watch out for

Outliers can distort the mean and variance — always pair a numeric summary with a **plot**.

Looking ahead

Ch. 2: **Relationships** between two variables — scatter plots and correlation.

Thank you!

Questions & Discussion

Next: Chapter 2 — Looking at Relationships